Domestic Cattle Optimization Algorithm

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Abstract. This paper proposes a novel swarm intelligence optimization algorithm, the Domestic Cattle Optimization Algorithm (DCOA), inspired by the natural behavioral characteristics of cattle in foraging, energy management, and group collaboration. The algorithm combines local fine-tuning, elastic jumping, leader guidance, energy metabolism and recovery, and local domain awareness to achieve adaptive search for complex optimization problems. The algorithm describes each update mechanism using purely mathematical formulas and provides detailed pseudocode. This paper focuses primarily on algorithm design and mathematical modeling and does not include experimental verification.

Keywords: Cattle optimization algorithm, swarm intelligence, metaheuristic algorithm, energy management, leadership mechanism.

1. Introduction

Swarm intelligence optimization algorithms are a type of optimization method inspired by natural biological behavior. These algorithms achieve global optimization of complex problems by simulating the collaboration, competition, foraging, and migration behaviors of biological groups in nature. Classic swarm intelligence algorithms include particle swarm optimization (PSO), ant colony optimization (ACO), bat algorithm, firefly algorithm, and wolf pack optimization algorithm. These algorithms demonstrate excellent search and convergence capabilities in dealing with high-dimensional, multimodal, nonlinear, and continuous optimization problems. The core idea of swarm intelligence optimization algorithms is to achieve collaborative optimization of individuals and the group through information sharing and adaptive behavior adjustment between individuals, thereby quickly approaching the global optimal solution in the search space [1-39].

As a docile and gregarious animal, domestic cattle exhibit a high degree of group collaboration and individual adaptability in their foraging behavior. In the natural environment, domestic cattle choose their movement paths based on their own energy levels, surrounding environmental conditions, and the behavior of other individuals in the group. Upon discovering high-quality forage resources, some individuals will lead others to that area, while individuals experiencing energy shortages or threats will opt

for localized fine-tuning or resting, thereby maintaining the overall group's survival and foraging efficiency. This herd behavior of cattle inspires the Domestic Cattle Optimization Algorithm (DCOA) proposed in this paper. Its core concept is to mimic the natural foraging behaviors of cattle, including foraging, energy management, local perception, and jumping, and transform these into a search strategy for the optimization algorithm.

Based on the natural foraging behavior of cattle, the proposed DCOA combines global and local search by representing individual positions as vectors X_i(t), the global optimal individual position as X_best(t), and the local optimal individual position as X_best_local(t). The algorithm's design combines global and local search to achieve efficient search for complex optimization problems. Each individual in the group dynamically updates its position based on its own energy E_i(t), local neighborhood fitness, group leader information, and potential jumping behaviors. DCOA is particularly well-suited for continuous optimization problems and has considerable scalability, allowing it to be applied to discrete or combinatorial optimization problems.

In the introduction, it's important to emphasize the advantages of swarm intelligence optimization algorithms and the unique value of cattle behavior simulation. First, swarm intelligence algorithms do not require gradient information and can handle non-smooth, discontinuous, or multimodal function optimization problems. Second, by simulating the natural foraging strategies of cattle, the algorithm achieves an adaptive balance between global search and local exploitation, avoiding the pitfalls of traditional optimization methods that can lead to local optima. Finally, the cattle optimization algorithm leverages individual energy management mechanisms and group leadership mechanisms to achieve greater information dissemination efficiency and search flexibility in complex search spaces. This design concept not only has theoretical value but also provides new solutions to practical engineering optimization problems.

2. Algorithm Inspiration and Behavior Modeling

2.1 Group Foraging Behavior

In natural environments, cattle typically search for optimal pasture areas in groups. Each individual moves dynamically based on its own energy status, information about the surrounding environment, and the positions of other individuals in the group. In the algorithm, individual positions are represented as vectors X_i(t), the individual with the best fitness in the group is denoted as X_best(t), and the local optimal individual is denoted as X_best_local(t). By simulating the foraging behavior of a herd of cattle, individuals combine local fine-tuning and global exploration within the search space, thereby achieving an efficient search strategy for optimization problems.

In the model design, each individual is influenced not only by its own historical experience but also by the guidance of the group leader and the surrounding environment at each iteration. Local fine-tuning corresponds to the individual's meticulous search of the local environment, while global exploration reflects the individual's ability to find optimal solutions across the entire search space. This foraging behavior of cattle provides a natural inspiration, enabling the algorithm to achieve good balance and adaptability in solving complex optimization problems.

2.2 Individual Energy Management

Cows consume energy while foraging. When energy is high, they can quickly explore new areas, while when energy is low, they tend to engage in local fine-tuning to conserve energy. To simulate this behavior, the algorithm introduces an energy variable $E_i(t)$ to represent the current state of the i-th individual and adjusts the individual's search strategy through dynamic energy updates:

$$E_i(t+1) = E_i(t) - consumption_rate * movement_distance$$

if distance_to_nearest_strong < threshold:

$$E_i(t+1) \mathrel{+}= recovery_factor * E_strong$$

$$E_i(t+1) = \min(E_i(t+1), E_max)$$

Where consumption_rate represents the energy consumption per unit distance moved, recovery_factor represents the energy recovery rate when approaching high-energy individuals, and E_max represents the upper limit of energy. Through this mechanism, individuals can adaptively adjust their exploration depth and movement range during the search process, achieving a dynamic balance between global exploration and local exploitation.

2.3 Group Leadership Mechanism

In natural groups, when some cattle discover areas of high-quality pasture, they will guide others there to improve overall foraging efficiency. In the algorithm, the top K individuals with the best fitness in the group are selected as leaders, and the leadership mechanism influences the positions of other individuals:

$$X_i(t+1) = X_i(t) + alpha * randn() * (X_best(t) - X_i(t))$$

$$+ delta * sum_over_K(leader_j - X_i(t)) / K$$

where alpha is the local fine-tuning strength, and delta is the leadership coefficient. This mechanism ensures that weak individuals can benefit from the experience of strong individuals, accelerating group convergence while improving information

dissemination efficiency, enabling the entire group to enhance collaborative capabilities in complex search spaces.

2.4 Elastic Jump Mechanism

When individuals are trapped in a local optimum, an elastic jump mechanism is introduced to prevent premature convergence. Jumping behavior is adaptively adjusted based on individual energy levels:

$$X_i(t+1) = X_i(t) + beta * (rand() - 0.5) * (U - L) * (E_i / E_max)$$

High-energy individuals can jump farther, expanding the search range, while low-energy individuals maintain smaller jumps to ensure local search accuracy. This mechanism enhances the algorithm's ability to escape local optima and provides a larger exploration space during global search.

2.5 Local Area Perception

During foraging, individuals can perceive the fitness of their surrounding neighborhood N_r to guide local fine-tuning:

$$F_local_mean = sum(F_j \text{ in } N_r) / |N_r|$$

if $F_local_mean < F_global_mean$:

$$X_i(t+1) = X_i(t) + alpha_local * randn() * (X_best_local - X_i(t))$$

This mechanism enables individuals to proactively fine-tune their positions based on their neighborhood environment, avoiding blindly following the global optimal solution. This improves the algorithm's adaptability and robustness for multi-modal and complex optimization problems.

3. Mathematical Model

3.1 Parameter and Variable Definitions

In the Domestic Cattle Optimization Algorithm (DCOA), the core parameters and variables involved in the algorithm must be clearly defined to accurately describe them in the mathematical model. Specific definitions are as follows:

D: Dimension of the problem, i.e., the number of decision variables in the optimization problem. The search space for each individual is a D-dimensional vector space, with the range of values for each dimension defined by L_d and U_d.

N: Population size, representing the number of individuals participating in the search in each generation. A larger N helps enhance global search capabilities but increases computational overhead.

T: Maximum number of iterations, representing the total number of rounds the algorithm attempts to update its position in the search space. The number of iterations should be appropriately set based on problem complexity and computational resources.

X_i(t): Position vector of the i-th individual in generation t, with each element corresponding to a decision variable in the optimization problem. The position vector represents the individual's current state in the search space.

F(X_i(t)): The objective function value of the i-th individual at the current position in generation t. This value measures the fitness or quality of the current position, and the optimization goal is to minimize or maximize this value.

E_i(t): The energy value of the i-th individual, which simulates the resource consumption of the individual during foraging or migration. High energy indicates that the individual is capable of extensive exploration, while low energy indicates a preference for local fine-tuning.

X_best(t): The global optimal position vector of the swarm in generation t, i.e., the position of the individual with the best objective function value in the current iteration.

In addition, auxiliary variables such as the local optimal position X_best_local(t), the jump vector triggered_jump_vector, and the leader set leader_j are involved to implement the algorithm's multi-layer search strategy and information transmission mechanism.

3.2 Initialization

In the initialization phase of the algorithm, each individual is assigned an initial position and energy, and the initial fitness is calculated to provide a reference for subsequent iterations:

for i = 1 to N:

X_i(0) = L + rand() * (U - L) # Randomly initialize the individual position in the search space

E_i(0) = E_max * rand() # Initialize the energy value to ensure individual state differentiation

F_i(0) = F(X_i(0)) # Calculate the initial fitness

X_best(0) = argmin(F_i(0)) # Select the initial global optimal individual

In practical applications, the randomness of initialization ensures the coverage of the group in the search space and avoids the search localization problem caused by the concentration of initial positions. The random initialization of E_i(0) can simulate the diversity of the energy states of the individuals in the group, making the search strategy more flexible in the early iterations.

3.3 Individual Position Update Formula

Individual position update is the core step of DCOA, combining three mechanisms: local fine-tuning, elastic jumps, and leader guidance. The formula is as follows:

X_i(t+1) = X_i(t)

+ alpha_local * randn() * (X_best_local - X_i(t)) # Local fine-tuning

+ beta_jump * triggered_jump_vector # Elastic jump

+ delta * sum_over_K(leader_j - X_i(t)) / K # Leader guidance

alpha_local: Local fine-tuning strength coefficient, used to control the movement of individuals during neighborhood search.

beta_jump: Jump strength coefficient, which determines the distance ratio that triggers a jump when local stagnation occurs.

triggered_jump_vector: Jump vector triggered by individual fitness history stagnation, which can expand the search space.

delta: Leader guidance coefficient, which measures the influence of the leader on individual position updates.

sum_over_K(leader_j - X_i(t)) / K: The average guidance vector of K leaders, allowing weak individuals to benefit from the experience of strong individuals in the group.

This formula embodies DCOA's multi-layered search strategy: local fine-tuning improves search accuracy, elastic jumping enhances global exploration capabilities, and leader guidance enables group information transfer and collaboration.

3.4 Energy Update Formula

The individual's energy state is dynamically adjusted in each iteration, simulating the foraging and resting behavior of domestic cattle in nature:

```
E_i(t+1) = E_i(t) - consumption_rate * movement_distance

if distance_to_nearest_strong < threshold:
E_i(t+1) += recovery_factor * E_strong
E_i(t+1) = min(E_i(t+1), E_max)
```

consumption_rate: The ratio of energy consumption per unit distance moved, reflecting the individual's metabolic expenditure during exploration.

movement_distance: The Euclidean distance between the individual's current position and the previous position.

recovery_factor: The energy recovery ratio when approaching a high-energy individual, simulating social energy sharing.

threshold: The distance threshold that controls when individuals begin to recover energy.

E_max: The energy limit, preventing excessive energy accumulation from causing excessive search jumps.

The energy update mechanism enables the algorithm to have adaptive search capabilities: high-energy individuals tend to conduct global exploration, while low-energy individuals focus on local fine-tuning, thus naturally balancing the exploration and exploitation ratio and preventing premature convergence of the group.

3.5 Global Optimization Update

The global optimal position is dynamically updated by comparing the current individual fitness with the group's global optimal value:

```
F_i(t+1) = F(X_i(t+1))
if F_i(t+1) < F(X_best(t)):
X_best(t+1) = X_i(t+1)
else:
X_best(t+1) = X_best(t)
```

This update strategy ensures that the global optimal solution is retained and propagated in each iteration, preventing high-quality solutions from being lost and providing a reference benchmark for the next generation of individual position updates.

4. Algorithm Flow

The iterative process of the cattle optimization algorithm can be described as follows:

```
Input: group size N, maximum number of iterations T, search space L, U, objective function F(X)
Initialize X_i(0), E_i(0)
Calculate F_i(0)
X_best(0) = argmin(F_i(0))

for t = 1 to T:
select top-K leaders # Select the leader with the best fitness
for i = 1 to N:
compute F_local_mean # Calculate the average fitness of the local neighborhood
if stuck_in_local_optimum:
trigger_jump() # Elastic jump mechanism
update X_i(t+1) using local, jump, leader guidance
update E_i(t+1) using consumption and recovery
update F_i(t+1)
update X_best(t+1)
Output X_best(T), F(X_best(T))
```

In each generation, the leader set is first determined. Then, all individuals update their positions and energy states based on local fine-tuning, elastic jumping, and leader guidance, and finally the global optimal solution. This process ensures efficient and adaptive exploration and exploitation in a complex search space.

5. Algorithm Features and Analysis

5.1 Multi-Layer Search Strategy

DCOA combines local fine-tuning with elastic jumping to achieve a balance between local search refinement and global search coverage. In complex multi-modal optimization problems, local fine-tuning ensures search accuracy, while elastic jumping prevents regression into local optima.

5.2 Dynamic Information Transfer

The leader mechanism rapidly transfers the experience of strong individuals to weaker individuals in the group, improving overall search efficiency and shortening convergence time. This information transfer is dynamic, and the leader is reselected at each iteration based on fitness to ensure group adaptability.

5.3 Adaptive Energy Management

The energy decay and recovery mechanism enables individuals to automatically adjust their exploration and exploitation ratios based on their own state. High-energy individuals tend to explore globally, while low-energy individuals prefer local search, naturally avoiding premature convergence and enhancing algorithm stability.

5.4 Local Perception

The local domain perception mechanism enables individuals to proactively fine-tune their positions based on neighborhood fitness, enabling intelligent local search. This mechanism is particularly important in multimodal optimization problems, improving the algorithm's adaptability to complex terrain and local optimal traps.

6. Discussion

The domestic cattle optimization algorithm (DCOA) is designed based entirely on natural behavior. It achieves a balance between global search and local exploitation through a leadership mechanism, energy management, elastic jumping, and local perception. The algorithm is not only applicable to continuous optimization problems but also has potential scalability and can be adapted for discrete or combinatorial optimization problems. This paper focuses on algorithm design and mathematical modeling, providing a theoretical foundation for subsequent experimental validation, parameter sensitivity analysis, and comparison with existing algorithms.

7. Conclusion

This paper proposes a new swarm intelligence optimization algorithm, the domestic cattle optimization algorithm (DCOA), which precisely describes individual position updates, energy management, elastic jumping, leadership, and local perception through mathematical formulas. The algorithm theoretically combines global and local search, providing a new approach to solving complex optimization problems. While this paper does not include an experimental component, it does provide a complete algorithm design and mathematical model, laying a solid foundation for future algorithm applications and experimental research.

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